

Cyber Threat Intelligence HackTheBox Exercise

Cyberthreat Intelligence

- Evidence-based knowledge of known or potential cyberthreats that allows organizations to proactively identify, prepare, prevent or mitigate cyberattacks.
- Gathers and analyze data from various sources, including open-source intelligence, dark web, threats feeds.

Threat Hunting

- Proactively finding potential threats and cybersecurity vulnerabilities in an organization network and system.
- Search for Indicators of Compromise and use threat hunting tools and platforms.

Threat Hunting Process:

1. Setting the stage:

- Planning and preparation phase.
- Threat hunting involves in-depth research of latest threat intelligence reports, analyzing industry-specific vulnerabilities, studying TTPs employed by threat actors.
- Tools like SIEM, EDR, IDS are configured to collect, correlate logs, generate alerts.
- Staying up-to-date with recent cyber threats.

2. Formulating Hypotheses:

- Making educated predictions about possible attacks based on various sources (threat intelligence, alerts, industry updates).

3. Designing the Hunt:

- After hypothesis, we develop hunting strategy.
- Recognizing data sources to analyze, tools to use, custom scripts or queries to create and IOCs that we will be hunting.

4. Gather and Examine Data:

- Gathering and analyzing by looking through logs and data using SIEM queries, EDR tools.

5. Evaluate Findings:

- Interpreting the results, looking for anomaly, threats.

6. Mitigate Threats:

- Taking remediation actions (isolating affected systems, eliminating malware, patching vulnerabilities, modifying configuration)

Threat Hunting Glossary

Adversary: Entity that is not authorized to access or modify information.

APT (Advanced Persistent Threat): APTs refers to cyberattacks where attackers gain and maintain unauthorized access to a target network to steal sensitive data, conduct espionage, or cause disruption. APTs are associated with highly organized groups or nation-state entities.

TTP:

- **Tactics:** Strategic objectives and high-level concepts of operation employed by adversary. It addresses “WHY” behind their actions.
- **Techniques:** The methods utilized by an adversary to accomplish their tactical objectives. “HOW” behind their actions.
- **Procedures:** Step-by-step instructions for implementation of each technique.

Threat:

- Event or potential situation that can cause harm to an organization’s operation, data, reputation, or assets by exploiting vulnerabilities in systems or networks.

Indicators of Compromise:

- Digital traces from active or past intrusions.
- IOCs can include hashes of malicious files, suspicious IP addresses, URLs, Domain names, malicious scripts.
- Analyzing IOCs can greatly enhance threat detection capabilities.
- This is what you put in your SIEM alerts.

Real SOC Analyst Workflow:

See suspicious activity → Check IOCs → Identify TTPs → Determine adversary type → Act.